

Selection Shaper

Separating what selection does *within* a generation from what actually gets passed to the next one — and finding the change the mean never reports.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/selection-shaper/

1 Before you open anything

Answer from what you already know. The point is to have something on paper to argue with later, not to be right now.

PREDICTION

A population of finches has a mean beak depth of **10.000 mm**. Deeper beaks crack the available seeds better, so the birds that survive to breed are not a random sample: they average **10.200 mm**. Predict the mean beak depth of their offspring generation.

Write one or two sentences saying how you got that number, and name anything you had to assume to get there.

2 Setting up a directional run

Open the demonstration. Three mode tabs sit above the graph; the sliders and the readout tiles are down the right-hand side (underneath the graph on a phone).

1. Set **Heritability h^2** to **0.50** first. This slider restarts the population at generation 0 every time you move it, so it goes first or it will undo your other settings.
2. Set **Strength of selection** to **0.40**.
3. Click the **Directional** tab.
4. Press **Reset**. The **Generation** tile should read 0.

5. Record the row for generation 0, then press **Step generation** once and record generation 1. Keep stepping to fill the rest.

Generation	Mean beak depth (mm)	Variance of beak depth (mm ²)	S – selection differential (mm)	R – response (mm)
0				
1				
2				
5				

a. The tile for S says *mean of the breeders minus mean of everybody*, and it names both numbers underneath. At generation 0, what do the breeders average, and what does the whole population average?

b. S and R are both measured in millimetres and both are non-zero here, but they are not the same number. Using the wording on the two tiles, say what each one is measured *across*.

c. The mean moves by the same amount every generation in this run, and the variance does not move at all. Read the **Variance of the breeders** tile and explain why the variance holding still is consistent with what that tile says.

3 What decides how much of S survives

Now change one thing only. Leave the mode on Directional and the strength on 0.40, and vary heritability.

1. Set **Heritability h^2** to **0.00**. The population restarts at generation 0 on its own.
2. With **Generation** reading 0, record S, R, and the **Predicted $h^2 \times S$** tile.
3. Repeat for each heritability in the table. Every value is reachable – the slider steps in units of 0.05.

h^2	S (mm)	R (mm)	Predicted $h^2 \times S$ (mm)	$R \div S$
0.00				
0.25				
0.50				
0.75				
1.00				

a. One column is identical in all five rows. Which one, and what does that tell you about whether heritability affects *who breeds*?

b. Write the rule connecting R, S and h^2 as an equation, then write it again as one sentence in your own words.

c. Go back to your prediction in Part 1. The breeders averaged 10.200 mm against a population mean of 10.000 mm — that is exactly the S you recorded. Which row of this table matches the answer you predicted? What did you assume h^2 was?

With $h^2 = 0$, selection is still working: the orange breeders curve is visibly to the right of the blue histogram and S is not zero. What is missing is heritable variation, so none of the advantage the breeders had is transmitted. Selection without heritable variation is not evolution.

4 Working backwards, and one place the shortcut fails

Backwards. A question is more likely to hand you the response and ask for the heritability than the other way round.

1. Set **Strength of selection** to **0.60**, mode **Directional**, and press **Reset**. Record S at generation 0: _____ mm.

- On paper, work out the heritability that would give a response of exactly **R = 0.150 mm** with that S. Show the working here.

- Set **Heritability h^2** to the value you calculated and read the R tile. Did it come out at 0.150 mm? If not, find your error before going on.

Where a reasonable shortcut breaks. “R = 0, so nothing is happening” is a natural thing to conclude. Test it.

- Set **Heritability h^2** to **0.50** and **Strength of selection** to **0.40**.
- Click the **Stabilizing** tab and press **Reset**.
- Record generation 0, then press **Step generation** until you have every row. The **h^2 right now** tile is the fourth one in *The two numbers to watch*.

Generation	Mean (mm)	S (mm)	R (mm)	Variance (mm ²)	Variance of the breeders (mm ²)	h^2 right now
0						
1						
3						
10						

- S is zero at every generation, and the fitness curve is nowhere near flat. Explain why S comes out zero here, in terms of where the green fitness curve peaks relative to the population.

- R is also zero, and the mean never leaves 10.000 mm. State one quantity in your table that *did* change, by how much between generation 0 and generation 10, and argue whether the population evolved.

c. The **h^2 right now** tile falls over the run even though you never touched the slider. Its subtitle shows the two numbers it is built from. Which of those two is dropping, and why does that make sense for selection that removes the extremes?

The population in this model is infinite, so nothing you see is sampling luck — every change is selection. There is also no mutation, so variance that stabilizing selection removes is gone for good and the run bottoms out at the non-heritable floor. In a real population mutation keeps feeding variance back in, and the two forces settle at a balance well above that floor.

5 Solve these without the demonstration

Close the page or look away from it. Show your working. Then reopen it, set the controls each question describes, and check yourself — if you were wrong, write down where it went off.

1. Directional selection, strength 0.40, heritability 0.75. At generation 0 the population mean is 10.000 mm and $S = 0.200$ mm. Find R , then predict the mean beak depth at generation 4. State the assumption about S you had to make to get from R to generation 4.

2. In a different run the population mean is 10.000 mm, the birds that breed average 10.300 mm, and the following generation averages 10.180 mm. Find S , find R , and find h^2 . Then say which mode and which two slider settings would reproduce this in the demonstration.

3. Over ten generations of stabilizing selection the mean beak depth is 10.000 mm at the start and 10.000 mm at the end. Someone concludes that the population is not evolving. Explain what they should be reading instead, and describe what has happened to the allele frequencies underlying beak depth. No calculation.

4. That same run starts with a phenotypic variance of 1.000 mm^2 and $h^2 = 0.50$. The variance falls each generation but never reaches zero, even after thirty generations. State the value it is approaching and explain, in terms of the two components of phenotypic variance, why selection cannot push it below that value in this model.

6 On your own

Nothing here is graded. The demonstration does considerably more than this worksheet used.

- Press **Disruptive, strong**, then press **Run** and press **Pause** at about generation 10. The mean stays at 10.000 mm while the histogram splits into two humps. Watch **Relative fitness of an average bird** as it happens, and work out why a value below 1.000 is exactly what disruptive selection means.
- Uncheck **Draw the predicted next generation**, set up any run you like, and sketch on paper where you think the dashed curve should sit relative to the orange breeders curve. Then switch it back on and see how close you were.
- Run **Disruptive, strong** until the distribution is clearly split, then click the **Directional** tab without pressing Reset. Compare the R tile against the **Predicted $h^2 \times S$** tile. $R = h^2S$ is exact only while the distribution is bell-shaped, and this one no longer is.